→ G2
→ Basmalamamdouh Abdelhamed

Q1- what is the OSI security architecture?

- security Attack :- any action that compromises the security of information owned by an organization
- security mechanism :- A process (or a device incorporating such a process) that is designed to detect, prevent, or recover from a security attack
- security service :- A process or communication service that enhances the security of the data processing system and information transfers of an organization

Q2- List and briefly categories of passive and active security?

Q3- Difference between passive and active security threats?

- Passive Attacks

The nature of passive attack is eavesdropping on, or monitoring of transmissions

Goal: To obtain information that is being transmitted

Types :-

- release of message contents

is easy understood A telephone conversation, an electronic mail message and a transferred file may contain sensitive or confidential info

- Traffic analysis :-

observing the pattern, location, frequency, length of message to guess the nature of the communication

- Active Attacks

involve some modification of the data stream or the creation of a false stream

- Goal :- To change data or affect the system's operation
- Types :-

mesquerade, Replay, modification of message, Denial of service (DoS)

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Q4- List and briefly define categories of security services?

security services :- it enhances security of data processing system and information transfers of an organization intended.

- X 800 defines it as :- a service provided by a protocol layer of communicating
- RFC 2828 defines it as :- processing or communication service provided by a system
- X.800 define it in 5 major categories:
  1- Anthentication : verifing identity
  2- Access control : preventing unauthorized access
  3- Data confidentiality : protecting data from unauthorized disclosure
  4- Data integrity : Ensuring data is not altered
  5- Non-Repudiation : preventing denial of sending/receving

Q5- List and briefly define of security mechanisms?

- Encryption
- Digital signature
- Authentication Exchange
- Routing control
- Access control
- Data integrity mechanisms
- Traffic padding
- Notarization

• specific security mechanisms

- Trusted functionality
- Event detection
- security recovery
- security labels
- security Audit Trail

• pervasive security mechanisms

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Problems :-

1.1 ATM system (PIN + card)

| Requirement | EX | importance |
|---|---|---|
| Confidentiality | → Protecting the PIN | → high |
| integrity | → Preventing balance modification | → high |
| Availability | → Ensuring the ATM is 'out of services' as rerely as possible | → high |

1.2 Telephone switching system

| Requirement | EX | importance |
|---|---|---|
| confidentiality | → Privacy of calls | → high / moderate |
| integrity | → correct call routing | → high |
| Availability | → continous service | → high |

1.3 Desktop Publishing system

a) confidentiality most important
(Government confidential reports)
b) integrity most important
(Legal or medical Books)
c) Availability most important.
(Daily newspaper Publishing system)

1.4 impact level (low / moderate / high)

| | C | I | A |
|---|---|---|---|
| 1) Public web server | Low | High | moderate |
| 2) Law Enforcement sentive information | High | High | moderate |
| 3) financial organization (Routine Administrative info) | Low | moderate | moderate |

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d) contracting organization

| | C | I | A |
|---|---|---|---|
| • sensitive constract data | High | High | moderate |
| • Routine administrative data | low | moderate | moderate |
| → system as whole → | High | High | High |

e) SCADA system (military power plant):

| | C | I | A |
|---|---|---|---|
| • Real Time sensor data | moderate | High | High |
| • Routine admin data | Low | moderate | moderate |
| → systam as whole → | High | High | High |

1.5 matrix between security service vs Attacks

| Attacks | Authentication | integrity | confidentiality | Avail |
|---|---|---|---|---|
| • Release of contents | X | X | ✓ | X |
| • Traffic analyss | X | X | ✓ | X |
| • masquerade | ✓ | X | X | X |
| • Replay | ✓ | ✓ | X | X |
| • modification | X | ✓ | X | X |
| • DoS | X | X | X | ✓ |

1.6 matrix between security mechanisms vs Attacks

| mechanisms | Release of contents | Traficanalysis | masqarade → |
|---|---|---|---|
| • Encryption | ✓ | X | X |
| • Traffic Padding | X | ✓ | X |
| • Authentication | X | X | ✓ |
| • Digital signature | X | X | ✓ |
| • Access control | X | X | ✓ |
| • Data integrity | X | X | X |
| • Routing control | X | X | X |
| • security Recovery | X | X | X |

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| Replay | Modification | DoS |
|---|---|---|
| ✗ | ✗ | ✗ |
| ✗ | ✗ | ✗ |
| ✓ | ✗ | ✗ |
| ✓ | ✓ | ✗ |
| ✗ | ✗ | ✗ |
| ✓ | ✓ | ✗ |
| ✗ | ✗ | ✓ |
| ✗ | ✗ | ✓ |

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